MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
FOURTH SEMESTER
Skill Enhancement Course (SEC) - MG4SECCSC201 - COMPUTERHARDWARE MAINTENANCE

SCHEME FOR EVALUATION

PART A: Very Short Answer (10 × 2 = 20 Marks)

Q1. Expansion of PCI and its use in a computer system. [K / CO1]

- PCI = Peripheral Component Interconnect.
- Used as a bus/slot on the motherboard to plug in expansion cards (network card, sound card, graphics card, etc.).

Marking:

- 1 mark – Correct expansion “Peripheral Component Interconnect” (or close spelling).
- 1 mark – Any correct use like: “to add extra cards,” “to connect devices like sound or LAN card,” “used for expansion slots.”

Q2. Device used to measure electrical current and one additional measurement. [K / CO1]

- Ammeter or Multimeter.
- Additional: voltage, resistance, continuity, etc.

Marking:

- 1 mark – “Multimeter” or “Ammeter” (accept “tester” if context clearly electrical).
- 1 mark – Any other valid quantity: voltage, resistance, continuity, etc.
- If they write “voltmeter” but say “measures current and voltage,” give at least 1 mark (liberal).

Q3. Expansion of BIOS and its use. [K / CO1]

- BIOS = Basic Input/Output System.
- Firmware that starts the computer, performs POST, and helps load the OS.

Marking:

- 1 mark – Correct expansion.
- 1 mark – Any reasonable use: “helps to start computer,” “checks hardware at startup,” “loads OS,” etc.

Q4. Define a USB port. [K / CO2]

- A standard interface/port used to connect peripherals (mouse, keyboard, pen drive, etc.) and transfer data/power.

Marking:

- 2 marks – Mentions “type of port/connector” + “used to connect devices like pen drive, keyboard, etc.”
- 1 mark – Says “place where we insert pen drive” or “port to plug devices” without “data transfer” idea.

Q5. Differentiate between remote troubleshooting and on-site troubleshooting. [U / CO2]

- Remote: problem solved using internet/phone, no physical presence.
- On-site: technician physically at the computer location.

Marking:

- 1 mark – Explains remote (online / from different place).
- 1 mark – Explains on-site (physically visiting).

Q6. Basic troubleshooting methods for keyboard or mouse errors. [U / CO2]

- Check cable/USB connection.
- Try different USB port.
- Restart system.
- Check if device is enabled in OS / drivers installed.
- Clean dirt/dust from mouse sensor/keys.
- Test with another keyboard/mouse.

Marking:

- 2 marks – Any **two sensible steps** listed.
- 1 mark – Only one valid troubleshooting step.

Q7. Define a malware signature. [K / CO3]

- A unique pattern or identifier (code, sequence) used by antivirus to detect specific malware.

Marking:

- 2 marks – Mentions “unique pattern/identifier” + “used by antivirus to find virus/malware.”
- 1 mark – Just says “a code or pattern of virus” without linking to detection.

Q8. Define expansion cards. [K / CO3]

- Circuit boards inserted into expansion slots (like PCI) to add extra functions: sound, network, video, etc.

Marking:

- 2 marks – Any description like “extra card plugged into motherboard to add functions” (sound card, graphics).
- 1 mark – “Hardware added to increase capacity” with no mention of slot or card.

Q9. List any two preventive maintenance activities for computers. [U / CO3]

- Regular cleaning (dust removal).
- Checking and securing cables.
- Updating antivirus / OS.
- Regular backups.
- Proper ventilation/cooling checks.

Marking:

- 2 marks – Any **two relevant activities**.
- 1 mark – Only one correct.

Q10. Mention two biometric techniques used to verify identity. [U / CO3]

- Fingerprint scanning.
- Iris/Retina scan.
- Face recognition.
- Voice recognition.

Marking:

- 2 marks – Any **two** valid biometric methods.
- 1 mark – Only one correct.

PART B: Short Answer (Any 6 × 5 = 30 Marks)

For each 5-mark question:

- 5 marks – Clear explanation + 3–5 good points or one solid paragraph + relevant example.
- 3–4 marks – Main concept clear, but fewer points or simple language.
- 1–2 marks – Only 1–2 relevant points, incomplete or vague.

Q11. Memory bank and its significance. [U / CO1]

- Memory bank: a group of memory slots or modules treated as a unit (on motherboard).
- Used to install RAM modules in matched groups.
- Significance: expands memory, matches bus width, improves performance.

Expected points:

1. Definition: group of memory chips/modules or slots (2 marks).

2. Helps in organizing RAM installation (1 mark).
3. Allows expansion of RAM (1 mark).
4. Supports speed/performance by matching system architecture (1 mark).

Q12. Necessity of peripheral devices. [U / CO1]

- Peripherals are needed for:
    - Input (keyboard/mouse).
    - Output (monitor/printer).
    - Storage (external drives).
- Without peripherals, user cannot interact with or see results from the system.

Expected points (any 4–5):

1. Definition: devices connected to computer externally.
2. Input: allow users to give commands/data.
3. Output: show results (monitor, printer).
4. Storage: additional space, portability (USB drives).
5. Communication: network devices (NIC, modem).

Q13. Distinguish between mouse and joystick. [U / CO1]

Core differences (IT Essentials style):

- Mouse:
    - Used for pointer control on screen (desktop work).
    - Movement mostly 2D on a surface.
- Joystick:
    - Used for games/simulations.
    - More 3D movement control (direction, angle).
- Buttons/usage examples.

Marking:

- 2 marks – Correct description of mouse.
- 2 marks – Correct description of joystick.
- 1 mark – At least one clear functional difference or example.

Q14. Role of error codes in diagnosing hardware failures. [An / CO2]

- Error codes (beep codes, POST codes, numeric codes, LEDs) indicate where problem lies (RAM, video, CPU).
- Save time and effort in diagnosis.
- Help technicians follow correct troubleshooting path.
- Reduce trial-and-error replacement.

Expected points:

1. Definition of error codes (2 marks).
2. Indicate specific faulty hardware area.

3. Reduce time for troubleshooting.
4. Prevent unnecessary part changes.
5. Example: beep codes during POST.

Q15. Analyze different types of backup strategies. [An / CO2]

- Full backup.
- Incremental backup.
- Differential backup.
- Sometimes mention: local vs cloud backup.

Expected points:

1. Full backup – all data (pros: simple; cons: time, space).
2. Incremental – only changed since last backup of any type.
3. Differential – changed since last full backup.
4. Need for backup strategies – protection against hardware failure, malware, loss.
5. At least one example (e.g., daily incremental + weekly full).

Q16. Role of diagnostic tools in troubleshooting. [U / CO2]

- Hardware diagnostics (memory test, HDD test).
- Software utilities (built-in OS tools, third-party tools).
- Purpose: detect, isolate, confirm faults.

Expected points:

1. Definition: programs/devices that check hardware health.
2. They test components like RAM, HDD, CPU temperatures.
3. Help identify exact cause of problem.
4. Reduce guesswork and save time.
5. Examples: memory test utilities, HDD SMART tools, etc.

Q17. Why high-performance computers require advanced cooling systems. [U / CO3]

- High-performance components consume more power → more heat.
- Excess heat causes throttling, errors, and damage.
- Advanced cooling (heat sinks, multiple fans, liquid cooling) needed to keep temperatures safe.

Expected points:

1. High-end CPU/GPU generate a lot of heat.
2. Heat affects performance (slow down / freezing).
3. Risk of permanent damage if overheated.
4. Need for better cooling designs (fans, airflow, liquid).
5. Example: gaming PCs, servers.

Q18. Guidelines to be followed during keyboard cleaning. [U / CO3]

- Turn off/unplug computer before cleaning.
- Remove loose dust using compressed air or brush.
- Use slightly damp (not wet) cloth; avoid liquid inside.
- Do not use harsh chemicals.
- Ensure everything is dry before reconnecting.

Expected points:

1. Power off/unplug for safety.
2. Use soft brush/compressed air.
3. Use mild cleaning liquid, not directly on keyboard.
4. Avoid water / strong chemicals entering keys.
5. Reconnect only when dry.